Understanding the BR-PB-TV Decomposition

An Intuitive Guide to Treatment Response in N-of-1 Trials

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Purpose. An introductory guide to the three-component decomposition of the symptom score into a drug response (BR), a placebo-belief response (PB) and a natural-course response (TV), as used by the package's data-generating process (DGP). It explains what each component represents, how the Hybrid design identifies it, and why the analysis model deliberately does not mirror the DGP. The worked examples are illustrative constructions, not simulation output.

Revision 2026-10-07. Reviewed against `analysis/report/NOTATION.md` and the current package source (`R/generateData.R`, `R/buildtrialdesign.R`, `R/utilities.R`). The Gompertz formula, its parameter interpretations and landmarks, the Hybrid design schedule and expectancies, the size of the joint covariance matrix, the description of participant heterogeneity, and the variance-of-a-sum identity were corrected; symbols were renamed to the canonical ones. Evidence labels used: *inspected* (read in the source), *derived*, *argued*, and *illustrative*.

Revision 2026-10-08. Two statements were corrected against `analysis/report/compendium-mathematics.t` the reset of t_{pb} at visits with $\eta_t = 0$ (Component 2), and the count of identified parameters in the phase-indicator extension, which is five, not six, because the full formula is exactly aliased (section "A small extension that usually pays off").

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined in this guide only.

Table 1: Notation index: symbols, meanings and status

Symbol	Meaning	Status
i, t	participant; visit (week)	canonical
Y_{it}	symptom score (code column `Sx`)	canonical
BL_i	baseline symptom level	canonical
BR_{it}, PB_{it}, TV_{it}	drug, placebo-belief and natural-course responses (reductions in symptoms)	canonical
ε_{it}	observation-level residual	canonical
u_i, v_i	participant random intercept; participant random slope	canonical; local
β_0, β_t	fixed intercept; fixed linear time slope	local
$D_{bc,it}$	exposure-decayed drug indicator, 1 on drug and $e^{-\lambda t_{sd}}$ off drug (code `Dbc`)	canonical
t_{od}, t_{sd}, t_{pb}	time on drug; time since discontinuation; expectancy-weighted time (code `tod`, `tsd`, `tpb`)	canonical
$t_{1/2}$, λ	carryover half-life; decay rate $\ln 2/t_{1/2}$	canonical
η	expectancy weight of a visit (code `e`, the `expectancies` argument): 1 open label, 0.5 blinded	canonical (paper 06, η(phase))
m, r, d	Gompertz maximum, rate and displacement (code `maxr`, `rate`, `disp`); subscripts BR, PB, TV name the component	local
σ_{PB}	open-label SD of the placebo-belief response	local
c_{cf1t}, c_{cfct}	cross-component correlations at the same visit and at different visits (code `c.cf1t`, `c.cfct`)	local
ρ, w_t	AR(1) serial correlation per week; calendar week of visit t	canonical; local
K	number of visits	local
I_ℓ	indicator of phase level ℓ (open label OL is the reference)	local
$\beta_{bm:D}$	biomarker-by-treatment interaction, the estimand (`bm:Dbc`)	canonical
$\beta_{bm}^{\mathrm{lumped}}$	biomarker slope of a one-component (lumped) regression	canonical
β_{bm}^{BR}, β_{bm}^{PB}, β_{bm}^{TV}	component-specific biomarker slopes	canonical
N	total number of participants across randomization paths	canonical

Glossary.

- **Component.** One of the three latent responses whose sum, subtracted from baseline, gives the symptom score.
- **Expectancy.** The design-fixed weight of the participant's belief that they are receiving active drug at a visit.
- **Open label (OL), blinded discontinuation (BDC), crossover (CO).** The three phases of the Hybrid design.
- **Identifiability.** Whether the data, under a given design, contain enough contrasts to estimate a component separately from the others.
- **Predictive / prognostic biomarker.** Associated with the size of the drug effect, or with the outcome irrespective of treatment.

The Big Picture: Why Split Responses?

Imagine you are a PTSD patient trying prazosin (a blood-pressure medication that also reduces nightmares). After eight weeks of treatment, your symptom score has dropped by twelve points. The clinical question is straightforward: 'is the drug working?' The statistical question is harder: 'how much of that twelve-point drop should I attribute to prazosin, and how much to other things?'

Three plausible mechanisms produce a drop in symptom score on a clinical trial:

1. **Drug effect.** The medication chemically modulates the physiology of nightmares. Prazosin is an alpha-1-adrenergic antagonist that reduces noradrenergic tone during REM sleep, and its pharmacological action is independent of whether you know you are taking it.
2. **Expectation effect.** You believe the treatment will work, and your belief activates well-characterized neurobiological pathways: dopaminergic reward signaling, descending pain modulation, anticipatory cortisol regulation. The placebo literature is now thirty years deep on this; it is not 'all in your head' in any dismissive sense.
3. **Natural history.** Some symptoms wax and wane on their own. PTSD severity drifts with anniversaries, life events, sleep hygiene, and the slow erosion that any chronic condition shows when measured repeatedly over months.

If we lump these three causes together as a single 'treatment response', we cannot distinguish them. The patient's twelve-point drop might be eight points of drug, three of expectation, and one of natural drift; or it might be two of drug, eight of expectation, and two of drift. The total is the same. The clinical implications are completely different.

The BR-PB-TV decomposition addresses this directly. It writes the observed change in outcome as a sum of three biologically interpretable components plus measurement noise:

$$Y_{it} \;=\; \mathrm{BL}_i - [\, BR_{it} + PB_{it} + TV_{it} \,] + \varepsilon_{it}.$$

Here BL_i is participant i's baseline symptom score, BR, PB, and TV are the three response components defined below (each a non-negative reduction in symptoms), and ε is observation-level noise. By decomposing the response into these three causes, we can ask cause-specific questions that the lumped form cannot answer: how much of the improvement is pharmacological, how much is expectation-driven, and how much would have occurred without any treatment at all.

The problem with one-component models

A naive analysis writes:

```
improvement = treatment_effect + error
```

The mathematical mistake is subtle but consequential. The 'error' term in such a model becomes a wastebasket for everything the analyst did not measure. If improvement is, in truth,

```
improvement = 4 (drug) + 6 (expectation) + 1 (natural history) + 1 (noise) = 12
```

the naive model fits this as

```
improvement = 12 (treatment_effect) + 0 (error)
```

at the population mean and infers that the drug is three times more powerful than it actually is. The structure is missed, and the inference is biased toward the lumped quantity rather than the quantity of clinical interest. Worse, the lumped quantity does not generalize: a population with stronger placebo expectations would appear to have a more powerful drug, and a population with milder natural-history drift would appear to have a less powerful one. Neither inference would be correct.

Decomposing the response separates these failure modes.

Component 1. Biological Response (BR)

What it is

BR is the **physiological response to the drug itself.** It is what happens because of the medication's chemical action on the body, independent of belief, suggestion, or natural history. A participant who somehow received prazosin without knowing it would still develop a measurable BR; a participant who took an inert sugar pill while believing it was prazosin would have a BR of zero no matter how strong their expectation.

Methodologically, BR is the quantity that drug regulators and prescribers actually want to know. Approval decisions, dose selection, and biomarker-guided patient selection are all framed around BR. The other two components are real and inferentially relevant, but they are not the substance of pharmacological efficacy.

Real example: PTSD and prazosin

Prazosin is an alpha-1-adrenergic antagonist originally developed for hypertension. It crosses the blood-brain barrier, and at the relevant doses it antagonizes noradrenergic activity in the locus coeruleus and prefrontal cortex during REM sleep. The mechanism for nightmare reduction is reasonably specific: prazosin damps the adrenergic surge that accompanies trauma-related dream content, and the resulting suppression of REM-stage arousal reduces nightmare frequency and severity.

The pharmacokinetic profile is short. Prazosin has a plasma half-life of about 2.3 hours (Hobbs et al. 1978); brain concentrations equilibrate quickly with plasma. This produces a characteristic on-drug / off-drug pattern:

- **Week 1 on drug.** Brain prazosin reaches steady-state within the first few doses. Nightmare scores begin to fall. The first- week reduction is typically small because the patient has not yet had many drug-on REM cycles to integrate.
- **Weeks 2-3 on drug.** Effect accumulates as the patient experiences several weeks of drug-on REM sleep. The reduction approaches its plateau.
- **Steady state.** Once the brain has fully adapted, additional weeks on drug produce essentially no further reduction. This is the saturating-curve property: the body has a ceiling for how much benefit any given drug can produce, and that ceiling is reached on a timescale set by the drug's own pharmacology and the disease's response time.
- **Discontinuation.** Within a few half-lives, plasma and brain concentrations return to negligible. BR drops to zero. The patient no longer has any pharmacological effect.

This pattern (saturating rise, plateau, sharp washout) is what BR captures. It is **not** linear in time; it is **not** symmetric on and off drug; and it is **not** modulated by what the patient believes. Those properties are the empirical signatures that let us identify BR separately from PB and TV.

Mathematical pattern: modified Gompertz

BR follows a saturating curve known as the modified Gompertz function, which has the form

$$BR(t_{od}) \;=\; m\,\frac{\exp(-d\,e^{-r t_{od}}) - e^{-d}}{1 - e^{-d}},$$

a Gompertz curve in time on drug t_{od}, shifted so that $BR(0) = 0$ and rescaled so that $BR(t_{od}) \to m$ as $t_{od} \to \infty$ (inspected: `modgompertz()` in `R/utilities.R`). The three parameters have direct interpretations:

Table 2: Modified Gompertz parameters for the BR component and their meanings

Parameter	Symbol	Meaning
Maximum response	m	The biological ceiling. The largest reduction the drug can produce, even with infinite time on treatment.
Onset rate	r	How fast the patient approaches the ceiling, given d. Higher r compresses the whole curve in time.
Displacement	d	The length of the initial lag. Large d gives a sigmoid curve with a slow start and an inflection at $t_{od} = \ln d / r$; small d (below 1) gives a curve that rises fastest at the start, with no lag.

For PTSD prazosin, the values used by Hendrickson et al. (2020) and throughout this compendium are $m = 10.99$ (about eleven points of symptom reduction at saturation), $d = 5$ and $r = 0.42$ per week. Their landmarks (derived from the

formula; `docs/40`, Section 4): the inflection is at 3.8 weeks, half of the maximum is reached at 4.7 weeks (about 33 days), and 90% at 9.2 weeks. This is a slow, sigmoid onset.

The Gompertz family was chosen because it captures the 'asymptote-from-below' shape that physiological dose-response curves typically display. Two alternatives that do **not** capture this shape:

- A linear function in t ($BR = \beta_t \cdot t$) climbs without bound and is biologically implausible past a few weeks.
- An exponential approach to the ceiling ($BR = m(1 - e^{-rt})$) is acceptable for fast-onset drugs but lacks the slow-start phase that characterizes most chronic-condition responses, where receptor adaptation, feedback regulation, and downstream gene-expression changes produce a measurable lag.

The modified Gompertz contains the exponential approach as its small-d limit (for small d, $\exp(-de^{-rt}) \approx 1 - de^{-rt}$, so the rescaled curve tends to $m(1 - e^{-rt})$; derived), and becomes an increasingly delayed sigmoid as d grows. Its three-parameter form is flexible enough to describe a range of pharmacodynamic profiles (argued; no fit to individual-level data is reported in this guide).

Why this matters for design

If we knew the participant's BR curve in isolation, we could read off the drug effect directly: BR at any timepoint is the answer to 'how much pharmacological reduction has accumulated by now?'. The trick of N-of-1 trial design is to construct contrasts between on-drug and off-drug timepoints that isolate BR from PB and TV. The blinded discontinuation phase, discussed in the trial-design section below, is precisely such a contrast: if the participant is taken off drug while still believing they are on drug, then BR collapses to zero while PB remains roughly intact. The difference between on-drug and off-drug response in that window is therefore a clean estimate of BR that does not require the analyst to model PB or TV explicitly.

Component 2. Placebo-Belief Response (PB)

What it is

PB is the change in symptom score that arises from the patient's **belief** that they are receiving treatment, independent of what they are actually receiving. This is the placebo response in the strict sense: the part of the improvement

that would persist if the active drug were silently replaced with an inert sugar pill, provided the patient continued to believe they were on the active drug.

It is essential to be precise about what PB is and is not. PB is not 'fake' improvement, and it is not 'all in the patient's head' in any dismissive sense. The placebo response is mediated by well- characterized neurobiological pathways. Belief-driven release of endogenous opioids and dopamine produces measurable analgesia and mood elevation. Anticipatory regulation of the hypothalamic-pituitary-adrenal axis modulates cortisol and inflammatory cytokines. Top-down attentional shifts change how symptoms are perceived and reported. Each of these mechanisms is a real biological response; what makes them 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

For trial design, the relevant point is that PB is **separately identifiable** from BR if and only if the trial design includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded discontinuation phase of the Hendrickson Hybrid design is the canonical such phase.

Real example: the blinded discontinuation contrast

Consider the Hendrickson Hybrid design as implemented in this compendium (inspected: `analysis/scripts/carryover-sensitivity/ simulation-core.R`), a twenty-week N-of-1 trial with visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, structured in three phases:

- **Weeks 1-8 (open-label on drug).** The patient knows they are receiving prazosin. Belief in treatment is at full strength (expectancy weight $\eta = 1$, in the model parameterization introduced below).
- **Weeks 9-12 (blinded discontinuation).** The patient is told that, at some unannounced point during this window, they may be silently switched to placebo. They cannot reliably tell whether they are still receiving the active drug. Belief in treatment drops to a partial level ($\eta = 0.5$) because the patient's expectation is now hedged. Everyone is on drug at week 9; the drug stops after week 9 in two of the four randomization paths and after week 10 in the other two.
- **Weeks 13-20 (blinded crossover).** Four weeks on drug and four on placebo, in randomized order. The crossover is also blinded, so $\eta = 0.5$ throughout it (`docs/40`, Section 11).

At week 10, a patient whose drug stopped after week 9 will have:

- $BR = 0$, or only its carryover residual (no pharmacology).
- $PB = 0.5 \cdot PB_{\text{open-label}}$, reflecting hedged belief.

- TV continuing on its slow trajectory.

At the same visit, a patient still on active drug will have:

- BR continuing at its near-plateau value.
- The same hedged $PB = 0.5 \cdot PB_{\text{open-label}}$ as the placebo-switched patient.
- The same TV trajectory.

The contrast between the two groups of paths at that visit is therefore a nearly clean estimate of BR, with PB matched (because both groups have the same belief) and TV matched (because TV is by construction independent of treatment). This is the inferential move that makes the BR-PB-TV decomposition identifiable from the data: the trial design supplies phases in which different combinations of components are present, and the analyst recovers the components by differencing those phases.

Mathematical pattern

PB also follows a modified Gompertz curve, with its own parameters, in the expectancy-weighted time t_{pb} (time accumulated over consecutive visits with positive expectancy; it resets to zero at any visit with $\eta_t = 0$), and its amplitude is scaled by the expectancy weight η_t of the current visit:

$$PB_t \;=\; \eta_t \, m_{PB} \, \frac{\exp(-d_{PB}\, e^{-r_{PB} t_{pb}}) - e^{-d_{PB}}}{1 - e^{-d_{PB}}}$$

(inspected: `buildSigma()` in `R/generateData.R` multiplies the Gompertz mean by the design's `e`). (Corrected 2026-10-08: this sentence previously described t_{pb} as time accumulated while the expectancy is positive. `buildtrialdesign()` in `R/buildtrialdesign.R` resets `tpb` to zero at every visit with `e = 0` and restarts the accumulation afterward (inspected), which matters only for designs with covert visits, such as a balanced-placebo design. See `analysis/report/compendium-mathematics.tex`, Section "Time arguments of the components, the shared clock, and the phase contrasts" (`sec:invb-clocks`).) Values of the expectancy weight:

Table 3: Expectancy weight η by trial phase with reasoning

Phase	η	Reasoning
Open-label on drug	1.0	Patient knows they are on the active drug.

Phase	η	Reasoning
Blinded discontinuation and blinded crossover	0.5	Patient believes they may or may not be on drug; expectation is hedged.
Open-label placebo	0.0-0.5	Patient told they are on placebo retains some residual expectation. Not used by any design in this compendium.

The choice of $\eta = 0.5$ rather than $\eta = 0$ for the blinded phases is itself a modeling assumption, inherited from Hendrickson et al. (2020). The empirical question is how much expectation a patient carries when uncertain about allocation. This guide does not cite a source for a specific range, and none was located in the project bibliography (flagged 2026-10-07; an earlier version quoted '30-60% of the open-label level' without a source). Setting $\eta = 0.5$ in the blinded phases is a convenient default, but sensitivity analyses should vary it to confirm that conclusions are not driven by this single number.

Why expectancy variance differs by phase

A subtle property of PB is that **its variance also scales with** η, not just its mean. When the expectancy factor is high (open-label on drug), there is substantial variability across patients in how strongly the placebo response is expressed: high- suggestibility patients may show large PB, low-suggestibility patients may show essentially none, and the population variance is correspondingly large. When η is low (blinded), the placebo response is uniformly attenuated, and the patient-to-patient variance shrinks accordingly. The model parameterization reflects this with a visit-dependent standard deviation (inspected: `buildSigma()` scales the PB SD by `e`):

$$\mathrm{SD}(PB_t) \;=\; \eta_t \, \sigma_{PB}.$$

This heteroscedastic structure matters for the analysis (argued): a working model with a common residual variance misstates the variability of the open-label visits (where PB contributes most variance) relative to the blinded ones, with consequences for both confidence intervals on the drug effect and the calibration of any subsequent biomarker test. The stress test of `docs/46` includes analyses with phase-specific residual variances for this reason.

Component 3. Time-Varying Response (TV)

What it is

TV is the change in symptom score that would have happened **even if the patient had received no treatment at all.** It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the course of the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

Like PB, TV is not 'noise'. It is a structured, participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

When does natural history actually exist?

A common simplifying assumption in trial analysis is 'subjects will improve over time, with or without treatment'. This is true for some conditions and dangerously wrong for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

Acute conditions tend to TV-positive. A cold, a sprained ankle, a post-operative pain syndrome, an episode of acute back pain: these conditions resolve, on average, on a timescale of days to weeks even without intervention. A trial of ibuprofen for tension headache or of a topical analgesic for ankle sprain will see substantial TV-improvement in the placebo arm, and the apparent drug effect must be measured *against* this baseline trajectory. Failing to model TV in an acute-condition trial is a common source of inflated effect estimates.

Chronic stable conditions tend toward TV-zero on the population mean, with substantial individual variance. Type 2 diabetes under stable management, treated hypertension, controlled chronic asthma, and long-standing PTSD without active triggers are all reasonably steady on the population mean across a sixteen-week trial. But individual participants show substantial variance: one patient is doing better this month because they have started sleeping more regularly, another is doing worse because their job has become more stressful. The population mean of TV may be near zero, but the participant-level TV variance is not, and the decomposition recovers both.

Chronic progressive conditions tend toward TV-negative. Alzheimer's disease and other neurodegenerative conditions, ALS, late-stage congestive heart failure, untreated progressive multiple sclerosis, and many forms of cancer all worsen on the timescale of months. A trial of a putatively neuroprotective intervention in early Alzheimer's must explicitly model the negative TV trajectory: the apparent treatment effect is the slowing of progression relative to the worsening that would otherwise occur. Treating TV as zero in such a trial would systematically under-estimate efficacy, because the placebo arm continues to worsen and the active arm's stability looks like 'no effect' rather than 'effect is preventing the natural decline'.

Cyclical and triggered conditions show structured TV that is neither monotone nor stationary. PTSD severity often spikes near trauma anniversaries; major depressive disorder has a measurable seasonal component; relapsing-remitting multiple sclerosis oscillates on a timescale of months. In these conditions, TV is not well-described by a monotonically increasing or decreasing Gompertz curve, and the model needs either a more flexible TV specification or an explicit covariate (e.g., season, days from nearest trigger) that absorbs the cyclic component before TV estimates the residual trend.

Selection-induced TV is real and rarely zero. Even a chronic stable condition can show TV-positive average trajectory in a trial population because of how participants are recruited. Patients enroll at a moment of clinical engagement, which often coincides with a recent worsening that triggered the help-seeking behavior. Regression to the mean alone (the statistical fact that an extreme observation tends to be followed by less extreme observations on average) can produce several points of apparent TV-improvement in the early weeks of a trial, even in a population whose underlying condition is stable. This is a recognized failure mode of single-arm trials in chronic-pain conditions: large 'treatment' effects can shrink when a control arm is added, because much of the apparent effect was regression to the mean (argued; see Barnett et al. 2005 for the mechanism).

Trial participation itself induces TV. The Hawthorne effect (participants improve simply because they are being observed), the structured-visit effect (regular visits with clinicians may improve symptom management directly), and the diary-completion effect (the act of monitoring symptoms changes how they are perceived and reported) are all real, all participant-mediated, and all confounded with calendar time. They appear in the data as TV-improvement that is independent of any pharmacological or placebo response, and they apply equally to all arms of the trial. Modeling them as part of TV is the appropriate handling.

Can we assume no TV effect?

The short answer is: rarely safely, and never without empirical support. The longer answer:

- **TV may be assumed zero only if** the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favor participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than to assume it is zero.
- **The decomposition makes the assumption testable.** A model that includes a TV component will return $\hat{m}_{TV}$ and a variance estimate; if both are small relative to BR and PB, the analyst has empirical evidence for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others.
- **Setting TV to zero and treating it as part of ε is worse than estimating it.** The participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug versus off-drug schedule. Lumping TV into noise produces residuals that are autocorrelated within participants and heteroscedastic across them, and the resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

The general principle: TV is a *structured* component of the response, not random noise, and the decomposition treats it accordingly.

Why this matters

If TV is ignored, every TV-induced point of symptom change gets attributed to either BR or PB or to noise. This is the most common sources of bias in chronic-condition trial inference (argued). A typical pattern: an uncontrolled trial that enrolls patients at peak symptom severity shows a large improvement that a concurrent natural-history arm would reveal to be largely regression to the mean; the patients would have improved similarly with no intervention (illustrative; an earlier version of this guide attributed this to 'early trials of chronic-fatigue therapies' without a source, and that attribution has been removed). TV in the model is the formal way to handle this risk.

The attribution problem is not symmetric. TV is positively correlated with the *opportunity* to detect a drug effect (chronic- condition patients are typically at peak severity at trial entry, and natural-history regression brings their scores down on average), and it is *negatively* correlated with the *credibility* of the detected effect (a regulator cannot tell, from a single-arm trial without a TV-control phase, how much of the observed improvement was treatment versus

regression). Both problems are addressed by decomposing TV explicitly and by including trial phases in which BR and PB are absent.

Mathematical pattern

TV also follows a modified Gompertz curve, in time since trial entry and parameterized independently of BR:

$$TV_t \;=\; m_{TV}\,\frac{\exp(-d_{TV}\,e^{-r_{TV}t}) - e^{-d_{TV}}}{1 - e^{-d_{TV}}}.$$

The maximum m_{TV} can be either positive (natural improvement) or negative (natural worsening). In the package's DGP the Gompertz parameters are population constants: the curve gives the mean of TV_t, and participants differ through a multivariate-normal deviation around that mean, with a common SD and AR(1) correlation across visits (inspected: `buildSigma()`). Hendrickson et al. gave TV the same parameters as PB ($m = 6.51$, $d = 5$, $r = 0.35$, SD 10; `docs/40`, Section 11), so the split between the two is an assumption rather than an estimate. A trial design that aggregates over many participants can identify the population mean trajectory and the participant-level variance, but only if there are enough visits to constrain the curve.

Why not just include 'week' as a covariate?

A natural objection is that a model with `week` as a covariate already controls for time. Three reasons this is not enough:

1. **Linearity assumption.** A `week` covariate enforces a linear trajectory: every additional week contributes the same fixed amount of natural-history change. Real disease trajectories are not linear; they curve, plateau, or accelerate. The Gompertz form is flexible enough to capture S-shaped or saturating patterns that linear time cannot.
2. **Participant-specific TV.** A single `week` coefficient estimates a population-mean trajectory, but participant heterogeneity in natural history is large and clinically relevant. The full BR-PB-TV decomposition uses participant- specific TV components, drawn from a population distribution, which is the correct way to express the heterogeneity.
3. **Confounding with BR onset.** Both BR and TV evolve smoothly over the same trial timescale. A `week` covariate cannot tell them apart from observational data alone; the design contrasts (on-drug vs. blinded vs. off-drug phases) are what supply identifiability. Without the explicit decomposition, the `week` coefficient absorbs both, and the analyst loses any ability to attribute the resulting time-trend.

For trials with rich within-person time structure (and N-of-1 trials are precisely such trials), the explicit Gompertz TV is the right tool, and the simpler `week` covariate is a coarse approximation that loses information.

How the trial design lets us identify the components

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. The Hendrickson Hybrid design, used as the worked example throughout the rest of this guide, has three phases that each contribute a different combination of components.

Table 4: Components present in each phase of the Hendrickson Hybrid design

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Blinded crossover	Either	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$

Three contrasts then bear on the three components:

- **Phase 1 vs. phase 2 (within drug):** the change from $PB(\eta = 1)$ to $PB(\eta = 0.5)$, about half of PB. This isolates the belief component only to the extent that BR and TV change little between week 8 and week 9.
- **Phase 2 or 3 on drug vs. placebo at the same visit:** BR alone (because PB and TV match across paths at a visit).
- **Off-drug visits:** $TV + PB(\eta = 0.5)$, plus any carryover residual of BR. Every off-drug visit is blinded, so TV is never observed without PB; and since the expectancy is positive at every visit, t_{pb} equals calendar time and the two components share one clock. TV and PB are separated only through the drop in expectancy at week 9 and their assumed curve shapes (inspected; `docs/40`, Section 3).

Each contrast targets a different component, and the analyst combines them through the linear-mixed-effects model fitted to the full trajectory. The model

is the formal mechanism, but the inference is driven by the design contrasts. In the Hybrid design the drug-versus-placebo contrast is the clean one; the TV versus PB split is weakly identified.

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

Worked example: what's lost when you don't decompose

Consider two participants, A and B, enrolled in a twenty-week N-of-1 trial of prazosin for PTSD. The example is illustrative: the numbers are chosen by hand to make the arithmetic transparent, and it simplifies the Hybrid schedule. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly.

- **Participant A** is a high placebo responder and a moderate pharmacological responder. Their true component values: $m^A_{BR} = 4$ (drug ceiling 4 points), $m^A_{PB} = 6$ (belief ceiling 6 points), $m^A_{TV} = 1$ (mild natural improvement).
- **Participant B** is a low placebo responder and a strong pharmacological responder. Their true component values: $m^B_{BR} = 8$, $m^B_{PB} = 1$, $m^B_{TV} = 1$.

In the open-label phase (weeks 1-8), both BR and PB run at full strength, and TV accumulates slowly. By week 8, both participants have reached close to their saturation values:

Table 5: Worked example components for participants A and B, open-label phase

	BR	$PB(\eta = 1)$	TV	Total improvement
Participant A	4.0	6.0	1.0	**11.0**
Participant B	8.0	1.0	1.0	**10.0**

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a

population drug effect of about ten points and would not distinguish the two participants.

Now consider the blinded discontinuation phase (weeks 9-12). Both participants are silently switched to placebo early in the phase (in the implemented design, after week 9 or week 10), and the expectancy weight drops from 1.0 to 0.5. By the end of week 12, the components are:

Table 6: Worked example components for participants A and B, blinded discontinuation phase

	BR	$PB(\eta = 0.5)$	TV	Total improvement
Participant A	0.0	3.0	1.0	**4.0**
Participant B	0.0	0.5	1.0	**1.5**

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits.

What does each modeling strategy infer?

One-component model. A fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B).
- Blinded-placebo means: 4.0 (A), 1.5 (B).
- Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75.
- Estimated placebo effect: not identifiable from this contrast.
- Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number (the drug effect, or rather the mean of (BR + the drop in PB induced by discontinuation)), conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B. It estimates 'drug' at 7.75 points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, and it has no principled way to decompose this further.

Three-component model. A linear-mixed-effects model with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. In the idealized case of noise-free data and a correctly specified model, the fit returns (illustrative, not a simulation result; with real data each estimate carries sampling error, and the TV versus PB split is weakly identified, as noted above):

- $\hat{m}^A_{BR} = 4.0$ (true 4.0). $\hat{m}^B_{BR} = 8.0$ (true 8.0). The drug effect is identified separately for each participant.
- $\hat{m}^A_{PB} = 6.0$ (true 6.0). $\hat{m}^B_{PB} = 1.0$ (true 1.0). The placebo response is identified, and the individual difference between A and B is preserved.
- $\hat{m}^A_{TV} = 1.0$, $\hat{m}^B_{TV} = 1.0$. Natural history is identified and matched, as it should be for participants who truly have similar trajectories.

What is at stake

Compare the inferences:

Table 7: One-component versus three-component answers to clinical questions

Question	One-component answer	Three-component answer
'Does the drug work for A?'	'Some response, magnitude unclear, ~7.75 points overall.'	'Yes: $m^A_{BR} = 4.0$ points, distinct from her larger placebo response.'
'Does the drug work for B?'	'Some response, magnitude unclear, ~7.75 points overall.'	'Yes: $m^B_{BR} = 8.0$ points, twice the magnitude of A.'
'Are A and B different responders?'	'No discernible difference; both improved by ~10 points.'	'Yes: B has 2x A's BR but 1/6 of A's PB.'
'If a biomarker predicts BR, is it useful here?'	Cannot answer (no BR estimate).	Yes if the biomarker correlates with $\hat{m}_{BR}$.
'Should A be continued on this drug long-term?'	Cannot say (the response could be largely placebo).	Probably yes (BR = 4 is real and pharmacological), but the larger placebo component will not persist if expectations decay.
'What is the population mean drug effect?'	7.75 points (biased; mixes BR and the discontinuation-induced drop in PB).	6.0 points = $(4.0 + 8.0)/2$ (correctly the mean of BR).

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. The information loss from lumping the components is not abstract: it makes the difference between concluding 'these patients respond similarly' and 'B has twice the pharmacological response that A does, and a biomarker predicting BR would distinguish them prospectively.' Predictive-biomarker validation, which is the

whole point of running an N-of-1 trial in this design family, becomes impossible without the decomposition.

A second example: the placebo-cohort fallacy

A subtler example shows the same issue at the population level. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug.

- **Cohort 1 (early trial, naive patients).** Mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$.
- **Cohort 2 (later trial, patients who have had multiple prior failed treatments and are skeptical).** Mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$.

By construction the drug works equally well in both cohorts ($BR = 5$ points). But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug 'works less well' in cohort 2 and might trigger a search for confounders, subpopulation effects, or methodological differences between the trials. None of this is real; the difference is entirely in PB, and specifically in the population baseline level of placebo responsiveness.

The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. The population difference is correctly attributed to PB, and the apparent drug-effect discrepancy disappears. This is the kind of reproducibility crisis that pharmacology has spent the last decade learning to navigate; the decomposition is one of the formal tools that makes it tractable.

Why the analysis model doesn't match the DGP

A reasonable reaction to the worked example is the following. 'If I am simulating data with three response components (BR, PB, TV) plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should I not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data?' The question is reasonable, and the standard answer is that four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and that a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

Four reasons not to match

1. The trial design must supply identifiability, and many designs supply only part of it. The BR-PB-TV decomposition is identified by phase contrasts: open-label versus blinded identifies the expectancy modulation of PB, on-drug versus blinded-placebo identifies BR, and off-drug timepoints identify TV. A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

2. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters: maximum, onset rate, displacement), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each typically with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\widehat{\beta}_{bm:D}$ is dominated by participant and timepoint counts; adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias. The linear-mixed model `Sx ~ bm + t + Dbc + bm:Dbc` with `corCAR1` residual correlation is, in this regime, the more powerful test for the question the analyst actually asks.

3. Component-matched analyses are non-linear in the parameters and convergence is fragile. Fitting Gompertz BR, PB, TV simultaneously requires either a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant N, the non-linear component-matched fit can be expected to fail to converge far more often than the linear analogue, and the non-converging replicates contaminate any aggregated analysis (argued; the convergence rate has not been measured in this compendium, and an earlier version's figure of 'below 90%' had no source).

4. The moderation estimand is robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions. They typically do not require a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question

whether the moderation is biological or expectation-mediated. A component-matched analysis would be required only for a secondary question of the form 'is this biomarker predicting the *pharmacological* response specifically, and not the expectation response?', and that secondary question typically admits a simpler answer through phase indicators rather than through full component fitting.

A small extension that usually pays off: phase indicators

The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),      correlation = corCAR1(...)
```

`phase` is a categorical covariate (open-label, blinded, post- crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR. (In the Hybrid design every open-label visit is on drug, so `Dbc:phase` and `bm:Dbc:phase` are identified only from the blinded phases and the open-label versus blinded on-drug difference; argued.)

With three phase levels, this extension adds six fixed-effect columns (two each for `phase`, `Dbc:phase` and `bm:Dbc:phase`) but only five identified parameters, within budget at realistic N. (Corrected 2026-10-08: this sentence previously counted six fixed-effect parameters. When every open-label visit is on drug, as in the Hybrid design, $D_{bc} = 1$ throughout the reference phase, so $D_{bc} = 1 - \sum_{\ell \neq \mathrm{OL}} I_\ell + \sum_{\ell \neq \mathrm{OL}} D_{bc} I_\ell$ exactly: the `Dbc` column is an exact linear combination of the intercept, the `phase` columns and the `Dbc:phase` columns, at every N and under any coding of the factor. The `bm:Dbc:phase` columns are not aliased. `nlme::lme` stops on the full formula with "Singularity in backsolve" (verified), and the Paper 06 driver, `phase_augmented_fit()` in `analysis/scripts/component-decomposition/00-common.R`, catches the error and refits without `Dbc:phase` (and skips the full formula for $N < 70$), so every reported phase-augmented fit is the reduced formula, recorded as `formula_dropped = 'Dbc:phase'` (inspected). Verified by the rank of the model matrix in `analysis/scripts/compendium-mathematics/check-gompertz-rmanova-cycles.R` (three-phase Hybrid: six columns added, rank gain five). See `analysis/report/compendium-mathematics.tex` Section "Rank of the phase-augmented model, and the drug-expectancy correlation by design" (`sec:invb-collinearity`).) It does not separately estimate BR and PB as components, but it does identify the *change* in apparent

drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to Gompertz parametric form.

When to actually fit the full decomposition

A component-matched analysis is worth the effort when:

- Sample size is in the several hundreds (a phase 3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials).
- The trial design provides timepoints in every phase combination (open-label on-drug, open-label off-drug, blinded on-drug, blinded off-drug, with enough density per cell to constrain the Gompertz curves).
- The scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point).
- A Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible and the analyst is prepared to defend the priors.

In that regime, `lcmm` (for component-specific class-based extensions), `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool, and the DGP and the analysis model deliberately do not match.

The asymmetry is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterize how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

The math behind it

The core model

For participant i at trial timepoint t:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; [\, BR_{it} + PB_{it} + TV_{it} \,] \;+\; \varepsilon_{it},$$

where the symptom score Y_{it} decreases as the components grow. Each component is a Gompertz function of its own time-on-state variable:

- BR_{it} depends on time-on-drug (cumulative exposure since drug initiation).
- PB_{it} depends on time-on-belief, modulated by the phase-specific expectancy factor η(phase).
- TV_{it} depends on time-since-trial-entry (calendar time).

Each component is a participant-specific random trajectory. In the package's DGP the Gompertz parameters (m, r, d) of each component are fixed population values that set the mean trajectory; each participant's trajectory is that mean plus a multivariate-normal deviation with AR(1) correlation over visits (inspected: `buildSigma()` in `R/generateData.R`). The parameters themselves are not drawn per participant.

Joint covariance structure

The components are not independent of each other. A patient with strong pharmacological response is, on average, also somewhat more likely to develop a strong placebo expectation ('this drug is clearly working, my belief in it strengthens'), and a patient with favorable natural history may be more likely to attribute that improvement to whatever treatment they happen to be on. The Hendrickson framework parameterizes this with two cross-component correlations:

- c_{cf1t}: correlation between different factors at a single timepoint. Captures the within-time covariance of BR, PB, and TV.
- c_{cfct}: correlation between different factors at different timepoints. Captures the across-time spillover (in the package it is further multiplied by $\rho^{|w_t - w_s|}$).

Together with the within-component AR(1) correlations, these define one joint covariance matrix per randomization path, of dimension $(3K + 2) \times (3K + 2)$: the biomarker, the baseline level, and K visits of each of TV, PB and BR, where K is the number of visits (26 variables in the eight-visit Hybrid design; inspected: the `labels` vector of `buildSigma()`). The simulation draws each participant's full trajectory from a single multivariate normal of that structure.

A common confusion: variance decomposition

A frequent slip is to write

$$\mathrm{Var}(Y) \;=\; \mathrm{Var}(BR) + \mathrm{Var}(PB) + \mathrm{Var}(TV) + \mathrm{Var}(\varepsilon)$$

as if this were an alternative to working with the full covariance matrix. By the bilinearity of covariance (the variance of a sum), the correct identity for any joint distribution is

$$\mathrm{Var}(BR + PB + TV + \varepsilon) \;=\; \sum_{c} \mathrm{Var}(c) \;+\; 2 \sum_{c<c'} \mathrm{Cov}(c, c'),$$

with the second sum over unordered pairs of distinct components (derived). So the simple sum-of-variances form holds when the components are pairwise uncorrelated, and more generally whenever the covariances sum to zero. Under the Hendrickson parameterization with c_{cf1t} and c_{cfct} both non-zero, the components are correlated, and the cross-covariance terms are non-negligible. Working with the full covariance matrix is therefore mandatory: the sum-of-variances shorthand is a special case that does not apply to the parameter regimes used in this project.

The practical consequence is that any variance-component analysis performed on simulation output should be done after the components are extracted (using the participant-specific posterior means or variance-component estimates from the fitted mixed model), not by reading off the marginal variance of Y. The marginal variance of Y is a single number that contains no information about the internal structure.

Common questions

Q: Could I just use a random intercept and random slope per participant?

A random intercept-and-slope mixed model is

$$Y_{it} \;=\; \beta_0 + u_i + (\beta_t + v_i)\, t + \varepsilon_{it},$$

with participant random intercept u_i and random slope v_i. This captures heterogeneity in baseline severity and in linear time trends, but it cannot decompose the time trend into BR-style saturating-drug, PB-style expectancy-modulated, and TV-style natural-history pieces. It also forces each participant's trend to be linear in t, which is biologically wrong for chronic-condition responses on the timescale of weeks. The BR-PB-TV decomposition is more specific about *why* the trajectory has the shape it does, and that specificity translates directly into identifiability of the clinically distinct components.

Q: Isn't TV just disease natural history? Why not measure it independently?

In principle yes. In practice, three obstacles:

1. **Ethical.** Withholding active treatment from patients with active symptoms to measure natural history is rarely acceptable.
2. **Practical.** Natural-history studies require long observation windows in the absence of intervention, and they typically lack the within-person matching that an N-of-1 design naturally provides.
3. **Statistical efficiency.** When the same trial collects on- drug and off-drug data on the same participants, the model extracts TV from the off-drug timepoints with high efficiency, and a separate natural-history study is redundant.

The decomposition is the way to recover TV from the trial itself, without paying the cost of a parallel natural-history study.

Q: Why is the biomarker only correlated with BR, not with PB?

This is a modeling assumption, not a fact about reality. The Hendrickson framework correlates the baseline biomarker (pretreatment standing systolic blood pressure in the motivating prazosin data; Raskind et al. 2016) with the BR factor only, and treats the biomarker as conditionally independent of PB and TV. The justification is that the biomarker is intended to be a *pharmacological* predictor: it measures something about the participant's likelihood of responding to the drug's mechanism, not their suggestibility or their natural-history trajectory.

If a candidate biomarker turned out to be correlated with PB, that would be a substantive finding (the biomarker is partly psychological in its action), and the model could be re-specified to allow that correlation. The package already offers it as an option (`c.bm.pb` in `buildSigma()`; inspected), and the stress test of `docs/46` adds prognostic couplings to the natural course and the baseline level as well (`docs/49`). The default specification reflects the prior that a useful predictive biomarker should be a clean pharmacological predictor; deviations from that prior are testable.

Q: What if TV is negative (the patient naturally worsens)?

In the standard parameterization with positive m_{TV}, the Gompertz form is monotonically increasing and TV is non-negative by construction. To allow naturally-worsening trajectories, the maximum m_{TV} can be drawn from a population distribution that includes negative values; the resulting Gompertz curve

is then monotonically decreasing in symptom-reduction (equivalently, monotonically increasing in symptom severity). The model handles this case without modification, and detecting $\hat{m}_{TV} < 0$ is itself informative: it implies the underlying condition is deteriorating, and the apparent treatment effect is partly camouflaging deterioration that would otherwise be visible.

Q: How many timepoints do I need to identify all three components?

Roughly, at least one timepoint per phase contrast that you want to exploit. A pure open-label design with eight weekly timepoints can identify $BR + PB$ jointly but cannot separate them. Adding a blinded discontinuation phase with at least three or four timepoints lets the model identify BR separately. Adding a longer off-drug follow-up extends identifiability of TV, particularly if the participant's natural-history trajectory is still evolving. The design contrasts table earlier in this document is the formal version of the answer: each component needs at least one phase in which it is uniquely present (or uniquely absent in a comparable arm), and the more such phases the better.

Q: What if my drug has a fast onset and a long washout?

The Gompertz parameters absorb this asymmetry. A drug with a fast BR onset has a high r; a drug with a long pharmacological washout has a long off-drug tail captured by the carryover half-life $t_{1/2}$ used in the analysis-side exposure-decayed drug indicator, $D_{bc,it} = 1$ on drug and $e^{-\lambda t_{sd}}$ off drug after exposure (0 before the first exposure; `docs/45`, Section 8). The decomposition is agnostic to the specific pharmacokinetics so long as the trial includes enough timepoints in the relevant transition windows (on-to-off and off-to-on) to constrain the curves.

Why this matters for biomarker validation

The motivating clinical question for the trial designs in this project is: 'does the biomarker predict treatment response?', and specifically 'does the biomarker predict the *pharmacological* component of treatment response?'. This is the precision-medicine question, and it depends critically on the BR-PB-TV decomposition.

The wrong way

A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If PB varies systematically across the biomarker distribution (because, say, the biomarker is correlated with optimism, suggestibility, or treatment-seeking history), then $\hat{\beta}_{bm}^{\text{lumped}}$ (the coefficient `beta_bm` above) contains a placebo-prediction component that is inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' by the biomarker would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

The right way

Regress *each component* on the biomarker:

```
BR    = beta_0_BR  + beta_bm_BR  * biomarker + error_BR
PB    = beta_0_PB  + beta_bm_PB  * biomarker + error_PB
TV    = beta_0_TV  + beta_bm_TV  * biomarker + error_TV
```

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker (it is partly psychological, or it correlates with natural-history characteristics), but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

The decomposition is therefore not optional for the precision- medicine question: it is the formal mechanism by which the question is asked at all.

Summary table

Table 8: Summary of the BR, PB, TV, and noise components and their roles

Component	Cause	Timescale	Identified by	Inferential role
BR	Drug pharmacology	Pharmacokinetic (hours-weeks)	Phase contrasts (on-drug vs. blinded placebo)	Pharmacological efficacy, biomarker-prediction target
PB	Belief in treatment	Expectancy-modulated (hours-weeks)	Open-label vs. blinded contrasts	Placebo response, accounted for, not the target
TV	Natural history	Disease-trajectory (weeks-months)	Off-drug timepoints	Confounding control
ε	Measurement noise	Per-observation	Residuals	Fit diagnostics

---

Key takeaway

The BR-PB-TV decomposition is not technical bookkeeping. It is the mechanism by which the trial design and the model together identify clinically distinct causes of symptom change from a single within-person trajectory. Without the decomposition, the analyst has only the lumped total, and the lumped total mixes pharmacological efficacy with placebo response and natural history. Pharmacological efficacy is the regulatory and clinical quantity of interest; the other two are biases that must be controlled for before efficacy can be inferred.

The trial design (open-label phase, blinded discontinuation, crossover) supplies the identifiability. The model extracts the components from the design contrasts. The biomarker validation question, if asked at the level of total response rather than the BR component, is unanswerable.

Further reading

- Hendrickson, Thomas, Schork, and Raskind (2020). *Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical*

methods and theoretical findings. Frontiers in Digital Health 2020;2:13. doi:10.3389/fdgth.2020.00013. PMID 34713026. The original paper that introduced this decomposition framework for N-of-1 trials targeting predictive biomarker validation.

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- Barnett AG, van der Pols JC, Dobson AJ (2005). *Regression to the mean: what it is and how to deal with it.* Int J Epidemiol 34(1):215-220. doi:10.1093/ije/dyh299. PMID 15333621.

Each entry above was checked against PubMed on 2026-10-07.

- Project documentation: `docs/06-ar1-residual-correlation.tex` for the within-factor serial-correlation structure; `docs/08-biomarker-correlation-decay.tex` for biomarker-BR correlation decay during off-drug periods; `docs/09a-carryover-bias-variance.md` for the bias-variance trade in carryover specification; `docs/19-mean-moderation-implementation-notes` for the mean-moderation architecture's implementation; `docs/40-hendrickson-hybrid-design-primer` for the Hybrid design, the published parameters, and the corrections applied to this guide in 2026-10; `analysis/report/01-dgp-mean-moderation-vs-mvn/` for the manuscript treatment of the BR moderation question.

Audience: graduate students in biostatistics, clinical research methodologists, and trialists encountering the BR-PB-TV decomposition for the first time. Familiarity with linear mixed models and Gompertz curves is assumed; deep knowledge of psychometric latent-variable methods is not.